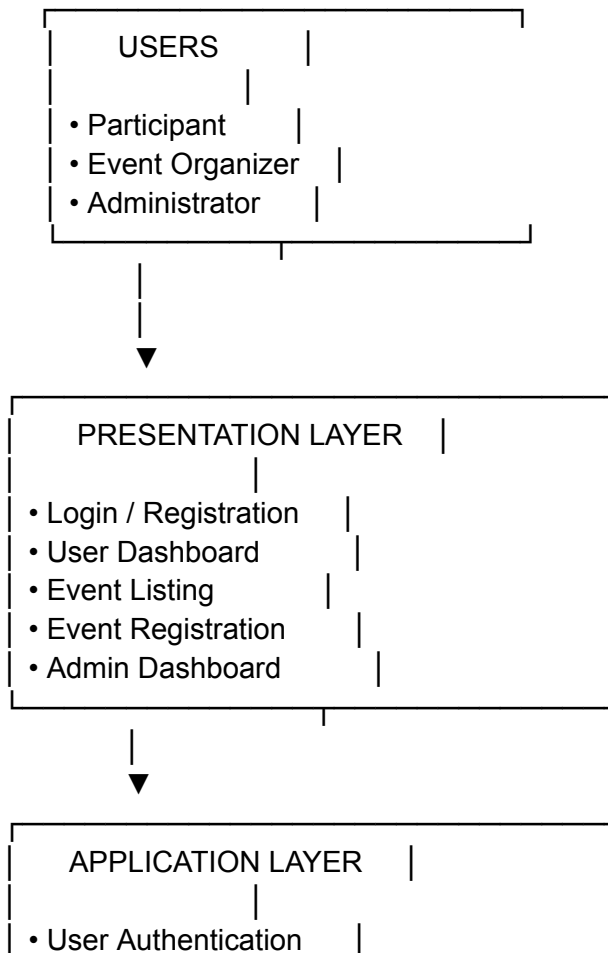


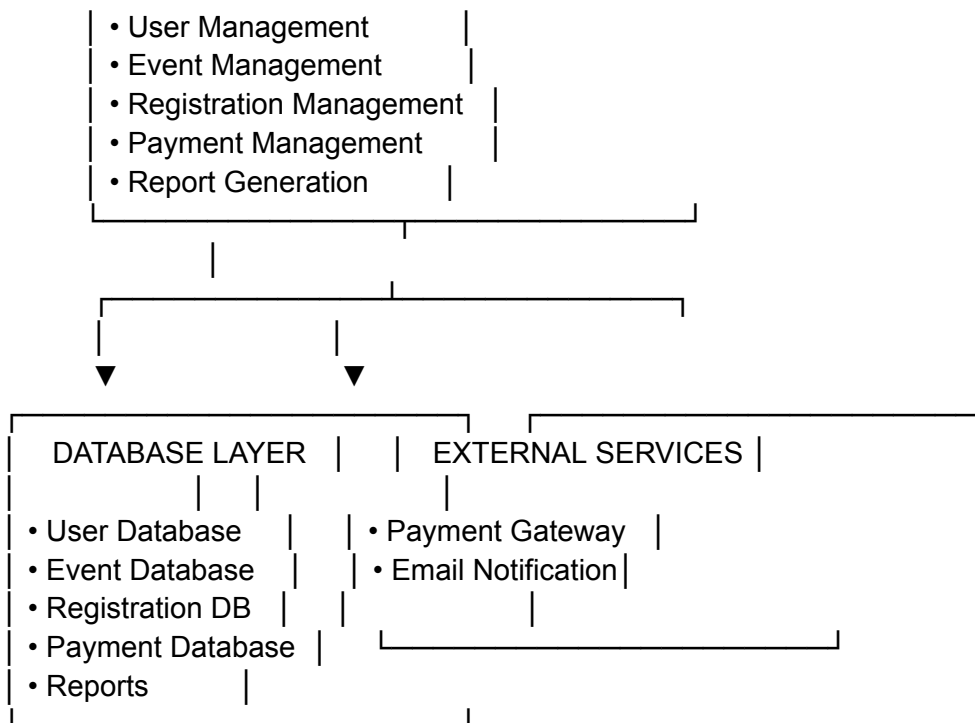
Project Design Phase

Solution Architecture

Details	Information
Date	15 February 2025
Team ID	_____
Project Name	Online Event Management System
Maximum Marks	4 Marks

1. Solution Architecture Diagram





2. Architecture Components

A. Presentation Layer

This is the interface through which users interact with the system.

- Registration and Login
- Participant Dashboard
- Event Listing
- Event Registration
- Organizer Dashboard
- Administrator Dashboard

B. Application Layer

This layer contains the main business logic of the system.

- User authentication and authorization
- User management
- Event creation and management
- Event registration
- Payment processing
- Participant management
- Report generation

C. Database Layer

The database stores and manages the information required by the system.

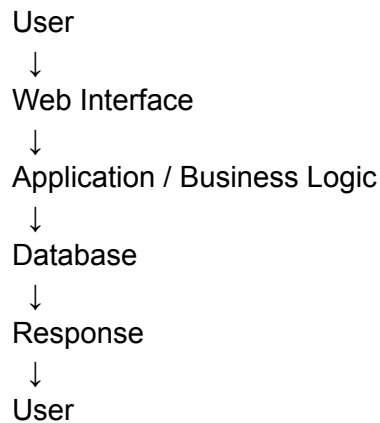
- **User Database** – participant, organizer and administrator details
- **Event Database** – event name, date, time, venue and description
- **Registration Database** – participant event registrations
- **Payment Database** – payment details and status
- **Reports** – event and registration information

D. External Services

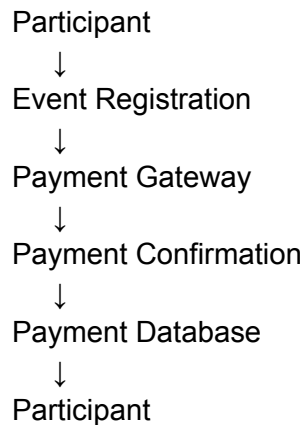
The system can communicate with external services such as:

- **Payment Gateway** – processes online event payments.
- **Email Notification Service** – sends registration and payment confirmations.

3. Data Flow



For payments:



4. Technology Stack

Layer	Proposed Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Web Server	Apache / WAMP
Payment	Payment Gateway API
Notifications	Email Service
Development Environment	Visual Studio Code / WAMP

5. Architecture Description

The **Online Event Management System** follows a **three-layer architecture** consisting of the Presentation Layer, Application Layer and Database Layer. Users interact with the system through the web interface. The application layer processes authentication, event management, registration, payment and reporting operations. The database layer stores user, event, registration and payment information.

The architecture separates the user interface, business logic and data storage, making the system easier to develop, maintain and expand. External services such as payment gateways and email notifications can be integrated to provide additional functionality.

6. Future Scalability

The system can later be extended with:

- Mobile application
- QR-code event check-in
- Digital certificates
- SMS notifications
- Advanced analytics
- Multiple payment gateways
- Social-media event sharing
- Cloud deployment